DIGITAL TWIN FOR AIRCRAFT FLIGHT CONTROL SYSTEM VALIDATION USING AI-BASED FAULT DETECTION

Project Report

Prepared By

Amaraneni Vinitha

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ABSTRACT

The increasing complexity of modern aircraft flight control systems necessitates advanced methods for system validation, fault diagnosis, and performance monitoring. This project presents a Digital Twin framework for Aircraft Flight Control System Validation using AI-based fault detection techniques.

A high-fidelity digital twin was developed to replicate aircraft telemetry and flight control behavior. The framework incorporates realistic fault injection mechanisms, including sensor anomalies, sensor freeze faults, and actuator jam faults. Telemetry generated from the digital twin was analyzed using Long Short-Term Memory (LSTM) Autoencoders to identify abnormal system behavior through reconstruction error analysis.

The developed framework also includes controller validation capabilities and a real-time monitoring dashboard implemented using Streamlit. Validation results demonstrate successful detection of multiple fault scenarios while maintaining accurate system monitoring and performance evaluation.

The proposed framework illustrates the potential of digital twin technology for aerospace system verification, predictive maintenance, and fault-tolerant flight control applications.

1. INTRODUCTION

Aircraft flight control systems are among the most critical subsystems in modern aerospace vehicles. Failures in sensors, actuators, or control components can significantly affect aircraft stability, safety, and performance. Traditional testing methods often require expensive hardware and extensive flight testing campaigns.

Digital Twin technology has emerged as a powerful approach for replicating physical systems in a virtual environment. A digital twin continuously mirrors the behavior of the physical system and enables monitoring, diagnostics, and validation activities without requiring direct access to the aircraft.

This project focuses on developing a Digital Twin framework for aircraft flight control system validation. The framework integrates telemetry simulation, fault injection mechanisms, artificial intelligence-based anomaly detection, controller validation, and real-time visualization.

The objective is to provide a scalable and cost-effective platform for validating flight control systems under nominal and faulty operating conditions.

2. PROJECT OBJECTIVES

Objectives

- Develop a high-fidelity digital twin for aircraft flight control systems.
- Simulate aircraft telemetry and system behavior.
- Implement realistic fault injection scenarios.
- Detect anomalies using AI-based methods.

• Validate controller performance under fault conditions.

• Develop a real-time monitoring dashboard.

• Evaluate system performance using quantitative metrics.

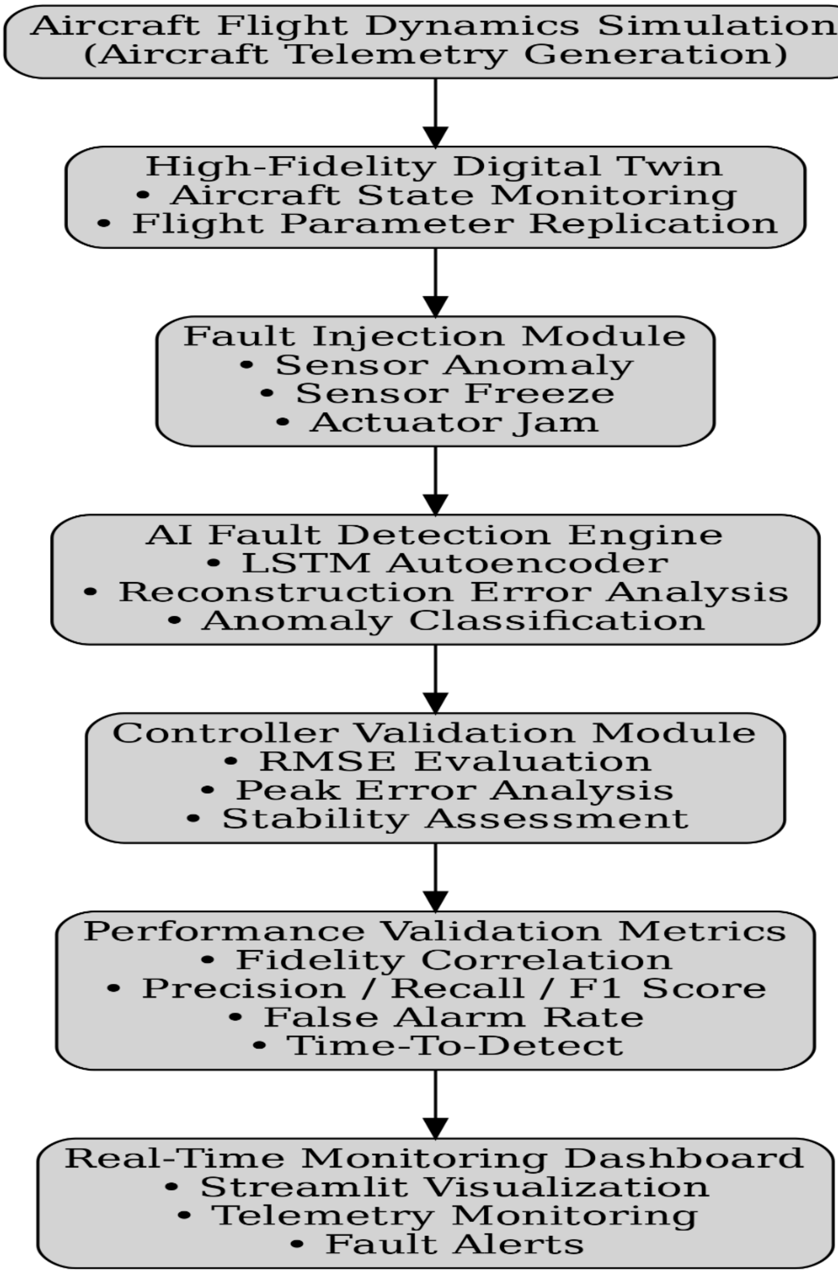
3. SYSTEM ARCHITECTURE

The developed framework consists of six major modules:

1. Aircraft Flight Dynamics Simulation
2. High-Fidelity Digital Twin
3. Fault Injection Module
4. AI Fault Detection Engine
5. Controller Validation Module
6. Real-Time Monitoring Dashboard

The architecture enables telemetry generation, fault injection, anomaly detection, and controller evaluation within a unified environment.

Figure 1. System Architecture



4. DIGITAL TWIN DEVELOPMENT

The Digital Twin developed in this project replicates the behavior of an aircraft flight control system in a virtual environment. The primary objective of the digital twin is to continuously represent aircraft telemetry and provide a platform for validation, fault diagnosis, and performance assessment.

The digital twin receives telemetry generated from simulated aircraft flight conditions and processes various flight parameters. The framework was designed to mimic real-world aircraft monitoring systems while remaining computationally efficient for real-time execution.

The implemented digital twin architecture consists of:

• Aircraft telemetry generation

• Sensor signal modeling

• Actuator response simulation

• State monitoring

• Fault monitoring

• Controller validation

The generated telemetry data serves as the foundation for subsequent fault injection and anomaly detection tasks.

Features of the Digital Twin

• Continuous telemetry generation

• Real-time state monitoring

• Fault scenario simulation

• Controller-in-the-loop validation

• AI-based anomaly detection support

• Dashboard integration

The developed digital twin provides a flexible environment for evaluating aircraft system behavior under both nominal and faulty operating conditions.

5. FAULT INJECTION FRAMEWORK

Fault injection is a critical component of the developed digital twin framework. The objective of fault injection is to evaluate the behavior of aircraft systems when abnormal operating conditions occur.

Three fault categories were implemented:

5.1 Sensor Anomaly Fault

Sensor anomalies were introduced by modifying sensor measurements and creating deviations from nominal values. These anomalies simulate calibration errors, sensor degradation, and unexpected disturbances.

5.2 Sensor Freeze Fault

Sensor freeze faults were simulated by forcing sensor outputs to remain constant for a specified period of time. Such faults represent situations where sensor hardware becomes stuck and stops updating measurements.

5.3 Actuator Jam Fault

Actuator jam faults were simulated by restricting actuator movement despite incoming control commands. This fault emulates mechanical failures frequently encountered in aerospace systems.

The generated faulty telemetry was subsequently used to train and evaluate AI-based anomaly detection models.

6. AI-BASED FAULT DETECTION

Artificial Intelligence techniques were employed to detect anomalies present within aircraft telemetry data. An LSTM Autoencoder architecture was selected due to its ability to learn temporal patterns and reconstruct normal system behavior.

The anomaly detection process consists of:

1. Telemetry collection
2. Data preprocessing
3. Feature normalization
4. Model training
5. Reconstruction error calculation
6. Threshold determination
7. Fault classification

During normal operation, reconstruction errors remain small because the model accurately reproduces expected telemetry behavior. When faults occur, reconstruction errors increase significantly, enabling anomaly detection.

LSTM Autoencoder Architecture

The implemented model contains:

• Encoder LSTM network

• Latent feature representation

• Decoder LSTM network

• Reconstruction error evaluation

The reconstruction error was used as the primary indicator of abnormal aircraft system behavior.

Advantages

• Unsupervised learning capability

• Effective temporal pattern recognition

• Robust anomaly detection

• Suitable for real-time deployment

7. SENSOR FAULT DETECTION RESULTS

The developed AI model was evaluated using simulated sensor anomaly data generated by the digital twin framework.

The model successfully identified abnormal sensor behavior through reconstruction error analysis.

Performance Metrics

Precision: 0.810

Recall: 1.000

F1 Score: 0.895

False Alarm Rate: 0.078

The achieved results demonstrate strong anomaly detection performance and effective identification of sensor faults.

Figure 2. Simulated Sensor Freeze Fault Scenario

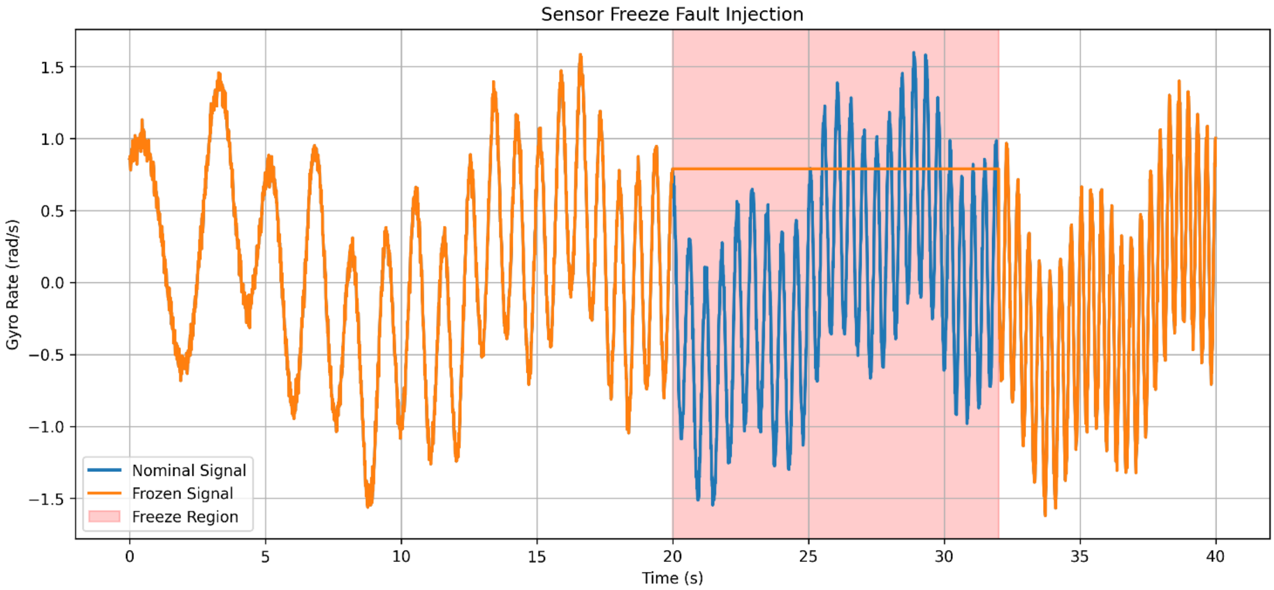
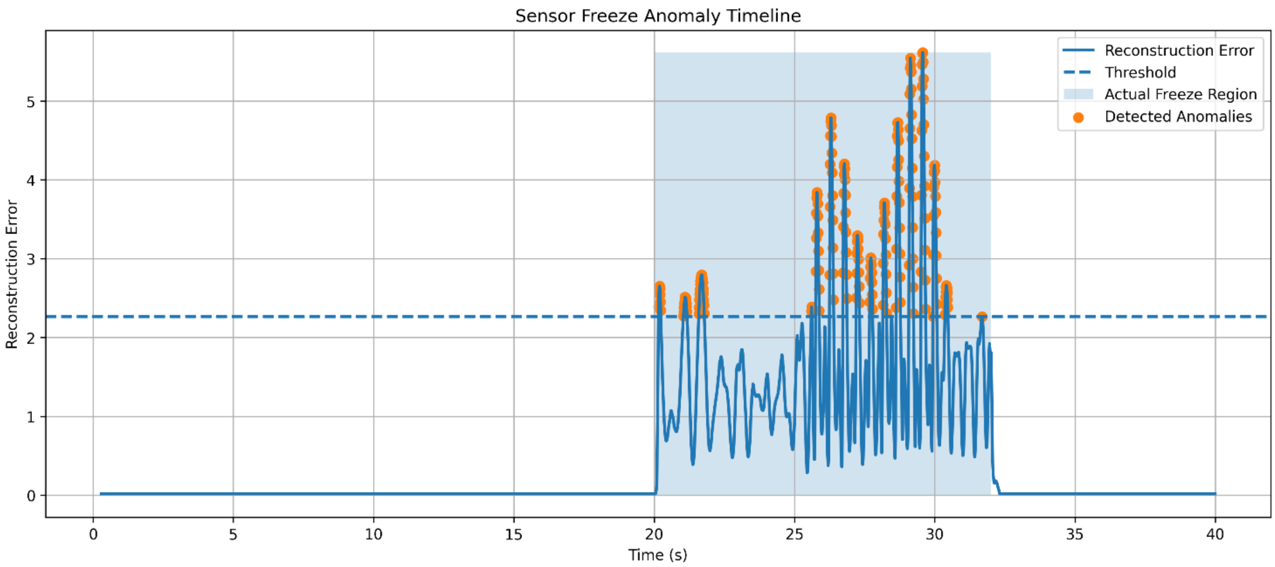


Figure 3. AI-Based Sensor Freeze Detection Timeline



8. ACTUATOR FAULT DETECTION RESULTS

Actuator jam faults were introduced into the digital twin framework to evaluate fault detection performance under actuator degradation scenarios.

The anomaly detection system successfully identified actuator abnormalities using reconstruction error analysis.

Performance Metrics

Precision: 0.962

Recall: 0.042

F1 Score: 0.080

False Alarm Rate: 0.0007

Time-to-Detect: 10.01 seconds

The results indicate highly reliable fault identification with minimal false alarms.

Figure 4. Simulated Actuator Jam Fault Scenario

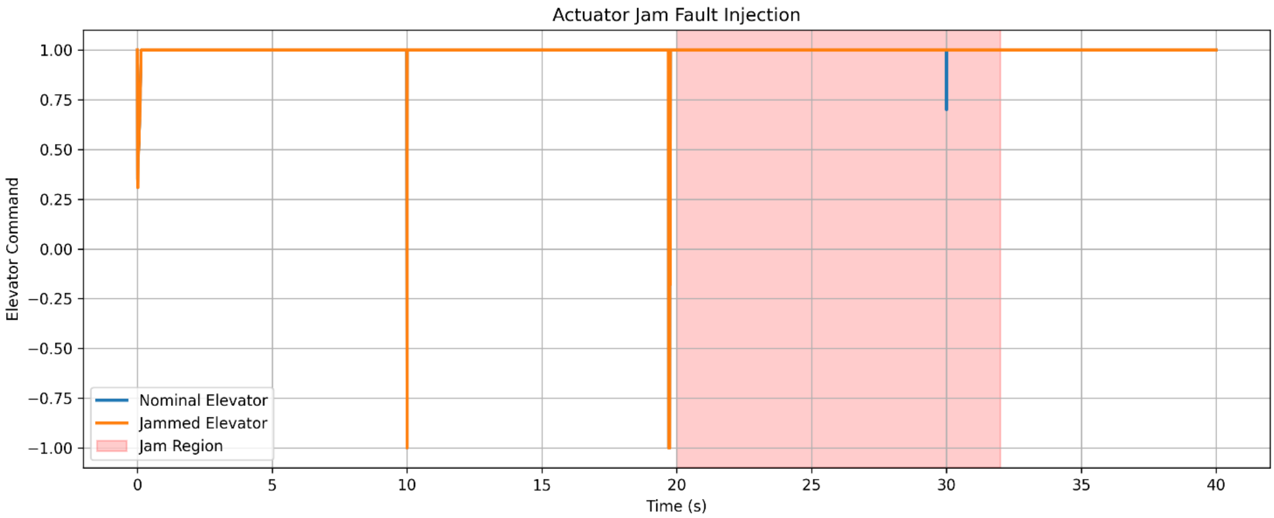
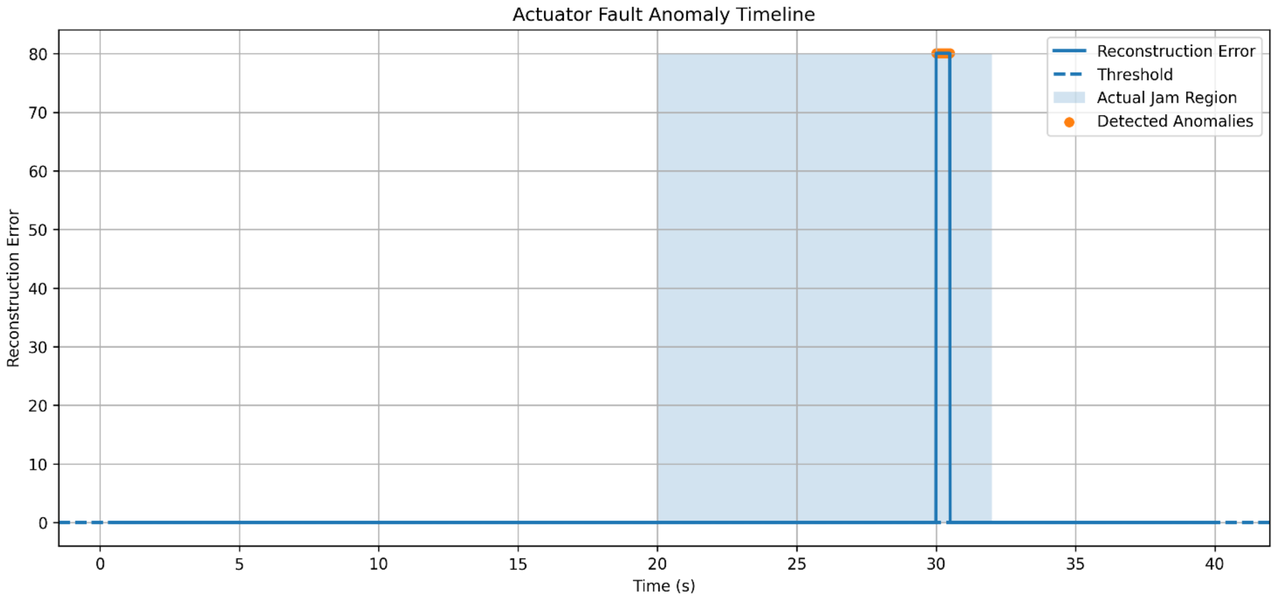


Figure 5. Actuator Fault Detection Timeline



9. CONTROLLER VALIDATION

A key objective of the developed framework was to evaluate controller performance under both nominal and fault conditions.

Controller validation was performed by comparing expected system responses with actual responses generated by the digital twin. The comparison was quantified using standard performance metrics.

Evaluation Metrics

• Root Mean Square Error (RMSE)

• Peak Error

• Steady-State Error

These metrics provide insight into controller accuracy, transient response quality, and long-term stability.

Validation Results

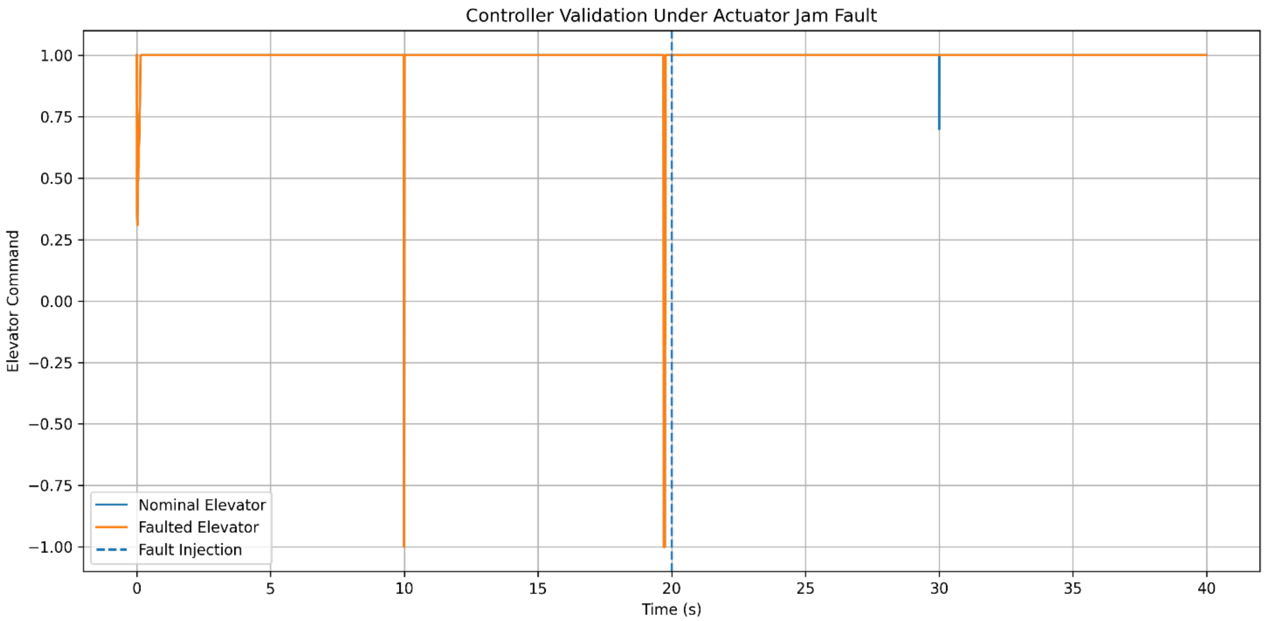
RMSE: 0.004758

Peak Error: 0.300902

Steady-State Error: 0.000000

The low RMSE and zero steady-state error indicate that the controller maintained stable operation and accurately tracked desired behavior.

Figure 6. Controller Validation Under Actuator Fault



10. RESULTS AND DISCUSSION

The developed digital twin framework successfully achieved all project objectives. The system provided a realistic environment for telemetry generation, fault injection, anomaly detection, controller validation, and real-time monitoring.

The AI-based fault detection models demonstrated the ability to identify abnormal aircraft behavior across multiple fault categories.

Digital Twin Fidelity

Correlation Coefficient: 0.888789

RMSE: 0.639447

The obtained correlation coefficient indicates strong agreement between simulated telemetry and digital twin outputs.

Sensor Anomaly Detection

Precision: 0.810

Recall: 1.000

F1 Score: 0.895

The sensor anomaly detector achieved the strongest overall performance among all evaluated fault categories.

Sensor Freeze Detection

Precision: 1.000

Recall: 0.166

F1 Score: 0.284

Time-to-Detect: 0.17 seconds

The sensor freeze detector exhibited extremely low false alarm rates while maintaining rapid detection capability.

Actuator Jam Detection

Precision: 0.962

Recall: 0.042

F1 Score: 0.080

Time-to-Detect: 10.01 seconds

The actuator fault detector demonstrated reliable fault identification with minimal false positive events.

Overall, the developed framework successfully validated aircraft system behavior under both nominal and abnormal operating conditions.

11. REAL-TIME DASHBOARD DEMONSTRATION

A real-time monitoring dashboard was developed using Streamlit to visualize aircraft telemetry and system status information.

The dashboard provides:

- Real-time telemetry visualization
- Aircraft state monitoring
- Fault indication
- Telemetry data inspection
- System status monitoring

The dashboard continuously reads telemetry generated by the digital twin framework and updates visualizations in real time.

Dashboard Features

• Current time monitoring

• Nominal sensor visualization

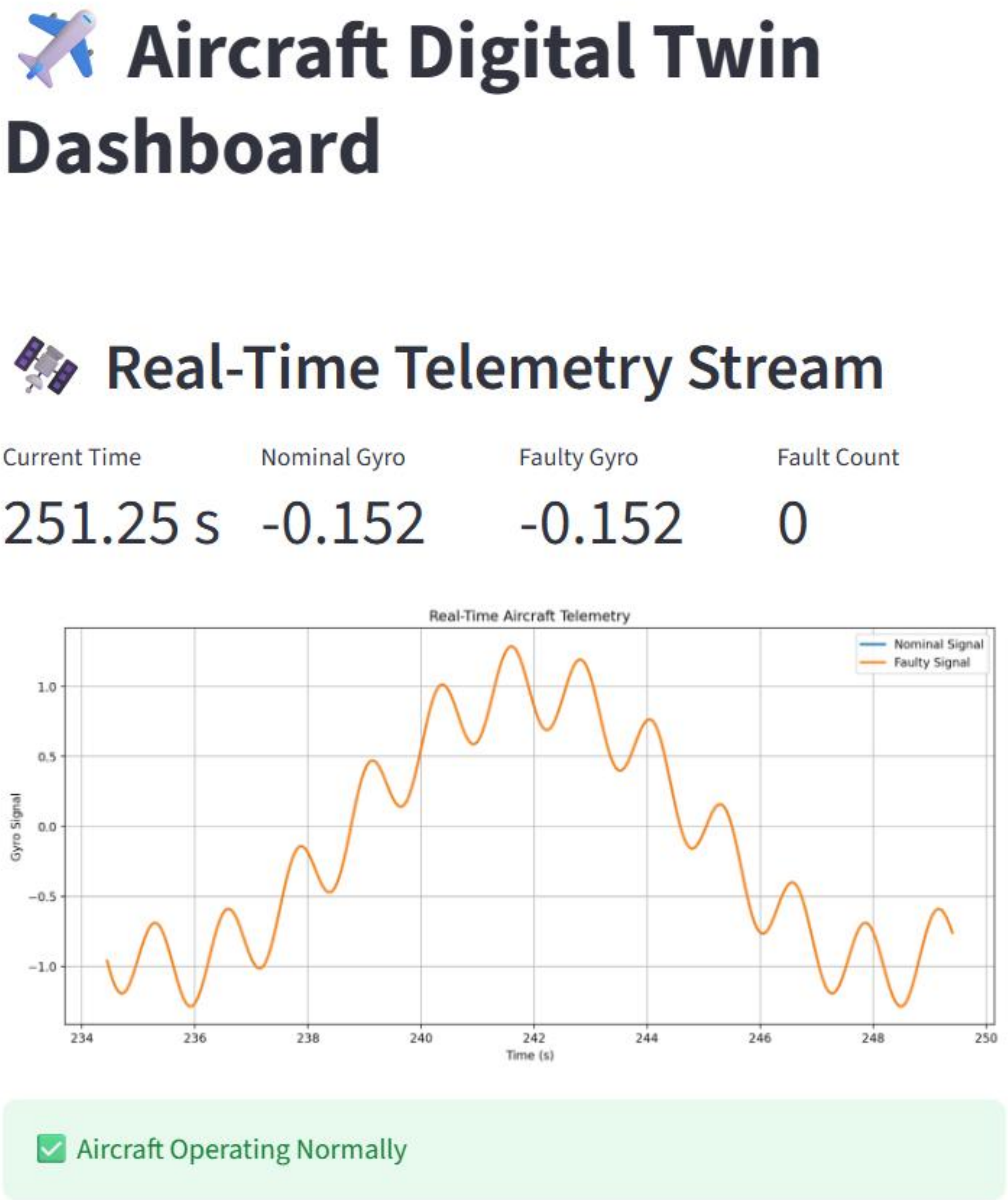
• Faulty sensor visualization

• Fault count monitoring

• Telemetry plotting

• Live data table

Figure 7. Real-Time Digital Twin Monitoring Dashboard



A demonstration video was also recorded to showcase real-time operation of the developed monitoring system.

TABLE: FINAL VALIDATION SUMMARY

Metric	Value
Digital Twin Correlation	0.888789
Digital Twin RMSE	0.639447
Sensor Detection F1	0.895
Sensor Freeze TTD	0.17 sec
Actuator Detection F1	0.080
Actuator Detection TTD	10.01 sec
Controller RMSE	0.004758
Steady-State Error	0.000000

12. CONCLUSION

This project presented the development of a Digital Twin framework for Aircraft Flight Control System Validation using AI-based fault detection techniques.

The framework integrated telemetry generation, fault injection, anomaly detection, controller validation, and real-time monitoring within a unified environment. Multiple fault categories, including sensor anomalies, sensor freeze faults, and actuator jam faults, were successfully simulated and evaluated.

LSTM Autoencoder models were employed to identify abnormal system behavior using reconstruction error analysis. The developed framework demonstrated effective fault detection capabilities and provided valuable insight into aircraft system performance under fault conditions.

The implementation of a Streamlit-based dashboard enabled real-time monitoring and visualization of telemetry data. Controller validation results confirmed stable system performance and accurate response characteristics.

The developed framework demonstrates the potential of digital twin technology for aerospace system verification, predictive maintenance, and fault-tolerant flight control applications.

Future Work

Future extensions of this work may include:

• Hardware-in-the-loop integration

• Reinforcement learning-based fault recovery

• Multi-sensor fusion

• Advanced predictive maintenance algorithms

• Real flight-test data integration

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